*The position of an object at any time t is given by*

$$s(t)=3t^4-40t^3+126t^2-9$$
$$s'(t)=12t^3-40(3)t^2+126(2)t$$
$$s'(t)=12t^3-120t^2+252t$$

*determine the velocity of the object at any time* $t: s'(t)$

*Does the object ever stop moving* $: s'(t)=0$ *?*

$$12t^3-120t^2+252t=0$$
$$t(12t^2-120t+252)=0$$
$$t=0, 12t^2-120t+252=0$$
$$t=7, t=3$$

when is the object moving to the right and when is the object moving to the left?

right/left movement → test with: $t=0.01\, t=-0.01, t=7.01, 6.98, t=3.01, 2.98$ etc

$$s'(0.01)=12(0.01)^3-120(0.01)^2+252(0.01)=2.508$$
$$s'(-0.01)=12(-0.01)^3-120(-0.01)^2+252(-0.01)=-2.53$$

$$s'(7.01)=12(7.01)^3-120(7.01)^2+252(7.01)=3.37$$
$$s'(6.98)=12(6.98)^3-120(6.98)^2+252(6.98)=-6.66$$

$$s'(3.01)=12(3.01)^3-120(3.01)^2+252(3.01)=-1.44$$
$$s'(2.98)=12(2.98)^3-120(2.98)^2+252(2.98)=2.87$$

right: $t<0 \vee (t>3 \wedge t<7)$
left: $(t>0 \wedge t<3) \vee (t>7)$